# Math 340: Elementary Matrix and Linear Algebra

# MIDTERM ONE

Wednesday, March 1st, 2023, 5:45pm-7:15pm

**Please circle your Instructor and your Discussion Session (date and start time given) in the table below.**

| Dr. Ruhui Jin | Dr. Yousheng Shi | Dr. Kaitlyn Phillipson | Dr. Jiuhua Hu |
|---|---|---|---|
| Marios, T 7:45 AM | Qi, M 8:50 AM | Joey, M 12:05 PM | Jingyi, T 11:00 AM |
| Ang, T 8:50 AM | Qi, M 9:55 AM | Sun Woo, M 1:20 PM | Jingyi, R 11:00 AM |
| Marios, R 7:45 AM | Zaidan, W 7:45 AM | Joey, W 12:05 PM | Gautam, T 12:05 PM |
| Ang, R 8:50 AM | Zaidan, W 8:50 AM | Sun Woo, W 1:20 PM | Gautam, R 12:05 PM |
| Ang, R 9:55 AM | Qi, W 9:55 AM | | Jingyi, T 9:55 AM |
| Emma, T 11:00 AM | | | |
| Emma, T 12:05 PM | | | |
| Marios, R 11:00 AM | | | |
| Emma, R 12:05 PM | | | |

Name: Key ______________________ Wisc email: ______________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of ten problems for a total of 50 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

**DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.**

1. (5 points, 1 point each) Are each of the following true or false statements? Circle your answer. Justification is not necessary.

invertible!

a.) **T** **F** If $A$ is a $5 \times 5$ matrix with $\det(A) = -7$, then the only solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.

b.) **T** **F** For any two $2 \times 2$ matrices $A$ and $B$, $(AB)^2 = A^2B^2$. AB may not equal BA.

c.) **T** **F** The only $3 \times 3$ matrix that is both symmetric and skew symmetric is the zero matrix.

d.) **T** **F** If $A$ is a $2 \times 2$ matrix and $\det(A) = -1$, then $\det(3A) = -3$. $\det(3A) = 3^2\det(A) = 9(-1) = -9.$

e.) **T** **F** If $A, B$ are $4 \times 4$ matrices with $\det(A) = 3$, and $B$ is obtained from $A$ by the row operation $2\mathbf{r}_2 + \mathbf{r}_1 \to \mathbf{r}_2$, then $\det(B) = 3$.

This is <u>not</u> an elementary row operation!

This is $2r_2 \to r_2$, then $r_2 + r_1 \to r_2$.

So $\det(B) = 2\det(A) = 6$.

2. (4 points) Let $A$ and $B$ be invertible (also called nonsingular) matrices with

$$A^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}, \ B^{-1} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 0 & 2 \\ 2 & -1 & 0 \end{bmatrix}.$$

Find the solution $\mathbf{x}$ to $AB\mathbf{x} = \mathbf{b}$, where

$$\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

(Hint: You do not need to compute $A$ and $B$ to solve this problem). Show all work.

$$AB\vec{x} = \vec{b} \implies \vec{x} = \underbrace{(AB)^{-1}}_{= B^{-1}A^{-1}}\vec{b}$$

so $$\vec{x} = B^{-1}A^{-1}\vec{b} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 0 & 2 \\ 2 & -1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}$$

$$= \begin{bmatrix} 2 & 0 & 3 \\ 0 & 0 & 2 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 18 \\ 12 \\ 0 \end{bmatrix}$$

3. (5 points total)

a. (2 points) Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a matrix transformation (also called a linear transformation) defined by $f(\mathbf{u}) = A\mathbf{u}$, where

$$\begin{array}{ccccc} 4\times 2 & & 2\times 1 & & 4\times 1 \\ A = \begin{bmatrix} 3 & 2 \\ 4 & -1 \\ 0 & 8 \\ 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} - \\ - \end{bmatrix} & & \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix} \end{array}$$

What's $n$? Answer: 2 (no justification necessary)

What's $m$? Answer: 4 (no justification necessary)

b. (3 points) Let $B$ be a $3 \times 2$ matrix and $C$ be a $p \times r$ matrix. If $(BC)^T$ is a $4 \times 3$ matrix, find $p$ and $r$. Show your work.

$BC \to 3\times r$

$3\times \underbrace{2\ p}_{=}\times r$ $p$ needs to be 2. $\boxed{p=2}$

$(BC)^T \to r\times 3 = 4\times 3$

$\boxed{r=4}$

4. (5 points) Let $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & x \\ 0 & 2 \end{bmatrix}$, and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$. Find $x$ (if possible) so that

$$(A+B)C = \begin{bmatrix} 3 & 9 \\ 5 & 4 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 5 & 2+x \\ 1 & 3 \end{bmatrix}$$

$$(A+B)C = \begin{bmatrix} 5 & 2+x \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -5+4+2x & 5+2+x \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 5 & 4 \end{bmatrix}$$

$-1+2x=3 \Rightarrow x=\frac{4}{2}=2$ ✓

<u>and</u> $7+x=9 \Rightarrow x=2$ ✓

$$\boxed{x=2}$$

5. (4 points)

Let $C = \begin{bmatrix} 2 & 0 & 2 \\ 1 & 1 & 0 \\ 0 & 2 & -3 \end{bmatrix}$. Compute $C^{-1}$. Show all work.

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & -3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{2}r_1 \to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & -3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{-r_1 + r_2 \to r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & 1 & 0 \\ 0 & 2 & -3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{-2r_2 + r_3 \to r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & -1 & 1 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{-r_3 \to r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array}\right] \xrightarrow{r_3 + r_2 \to r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & -\frac{3}{2} & 3 & -1 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array}\right]$$

$$\xrightarrow{-r_3 + r_1 \to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & -2 & 1 \\ 0 & 1 & 0 & -\frac{3}{2} & 3 & -1 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array}\right]$$

$$\boxed{C^{-1} = \begin{bmatrix} \frac{3}{2} & -2 & 1 \\ -\frac{3}{2} & 3 & -1 \\ -1 & 2 & -1 \end{bmatrix}}$$

6. (3 points) Suppose $\mathbf{a} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Write $\mathbf{c} = \begin{bmatrix} 3 \\ -12 \end{bmatrix}$ as a linear combination of $\mathbf{a}$ and $\mathbf{b}$.

$$x \begin{bmatrix} 2 \\ -4 \end{bmatrix} + y \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -12 \end{bmatrix}$$

$$2x = 3 \rightarrow x = 3/2$$

$$-4x + 3y = -12$$

$$-4(3/2) + 3y = -12$$

$$-6 + 3y = -12$$

$$y = \frac{-6}{3} = -2$$

$$3/2 \begin{bmatrix} 2 \\ -4 \end{bmatrix} + -2 \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -12 \end{bmatrix}$$

7. (4 points) Let $D$ be the $4 \times 4$ matrix obtained from $E$ after the following elementary row operations:

$10\mathbf{r}_1 + \mathbf{r}_3 \to \mathbf{r}_3$

$\frac{3}{2}\mathbf{r}_3 + \mathbf{r}_4 \to \mathbf{r}_4$

$\mathbf{r}_2 \longleftrightarrow \mathbf{r}_3$

$k\mathbf{r}_2 \to \mathbf{r}_2$

$(-4)\mathbf{r}_2 + \mathbf{r}_3 \to \mathbf{r}_3$

$\frac{1}{4}\mathbf{r}_4 \to \mathbf{r}_4.$

If $\det(D) = 12$, and $\det(E) = 10$, find $k$. Show all work.

$$E \xrightarrow[\text{no change}]{10r_1+r_3\to r_3} E_1 \xrightarrow[\text{no change}]{\frac{3}{2}r_3+r_4\to r_4} E_2 \xrightarrow[\text{neg.}]{r_2\leftrightarrow r_3} E_3 \xrightarrow[\times k]{kr_2\to r_2} E_4 \xrightarrow[\text{no change}]{-4r_2+r_3\to r_3} E_5 \xrightarrow{\frac{1}{4}r_4\to r_4} E_6 = D$$

$$\text{det: } 10 \longrightarrow 10 \longrightarrow 10 \xrightarrow{\text{neg.}} -10 \xrightarrow{\times k} -10k \longrightarrow -10k \xrightarrow{\times\frac{1}{4}} -\frac{10}{4}k = \det(D) = 12$$

$$12 = -\frac{10k}{4} \qquad k = -\frac{12 \cdot \cancel{4}^{2}}{5\cancel{10}} = \boxed{\frac{-24}{5}}$$

8. (7 points total)

(This problem has overflow room on next page if needed)

Suppose $A$ is a $3 \times 3$ matrix with $A^3 = \begin{bmatrix} 0 & 1 & 3 \\ 3 & 2 & -5 \\ 1 & 1 & 2 \end{bmatrix}$. Note that this matrix is $A^3 = AAA$, not $A$.

a. (4 points) Find $\det(A)$. Show all work.

↳ $\det(A^3) = (\det(A))^3 = -8 \Rightarrow \boxed{\det(A) = -2}$

$$\det(A^3) = \det \begin{bmatrix} 0 & 1 & 3 \\ 3 & 2 & -5 \\ 1 & 1 & 2 \end{bmatrix} = 0 \cdot (\text{w}) - 1(3 \cdot 2 + 5) + 3(3 \cdot 1 - 2)$$

$$= -11 + 3 = -8.$$

b. (3 points) Consider the matrix transformation (also called a linear transformation) $f(\mathbf{u}) = A\mathbf{u}$.

Is $\mathbf{w} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ in the range/image of $f$? Fully justify your answer.

Want to know: if there is $\vec{u}$ w/ $A\vec{u} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$\det(A) \neq 0 \Rightarrow A$ is invertible!

so $A\vec{u} = \vec{w}$ must have a solution

(it's $\vec{u} = A^{-1}\vec{w}$).

(Overflow room for Problem 8, if needed)

Alternate (less optimal) solution: show $A^3\vec{u} = \vec{\omega}$ has a soln:

$$\left[\begin{array}{ccc|c} 0 & 1 & 3 & 3 \\ 3 & 2 & -5 & 2 \\ 1 & 1 & 2 & 1 \end{array}\right] \xrightarrow{\text{math...}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -9/4 \\ 0 & 1 & 0 & 15/4 \\ 0 & 0 & 1 & -1/4 \end{array}\right] \rightarrow \vec{u} = \begin{bmatrix} -9/4 \\ 15/4 \\ -1/4 \end{bmatrix} \rightarrow A^3\vec{u} = \vec{\omega}.$$

If $A^3\vec{u} = \vec{\omega}$, then $A(\underbrace{A^2\vec{u}}_{\vec{x} = A^2\vec{u}}) = \vec{\omega}$.

so $A\vec{x} = \vec{\omega}$ has a solution.

9. (7 points total) Suppose a linear system of 3 equations and 3 unknowns $(x, y, z)$ reduces to the following form:

$$\begin{array}{c} ① \\ ② \\ ③ \end{array}\left[\begin{array}{ccc|c} 1 & 0 & 4 & 3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & (a-3) & a+5 \end{array}\right]$$

a. (2 points) For which value(s) of $a$ (if any) does this system have **no solutions**? Briefly explain your answer.

If $a=3$: we get $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 8 \end{array}\right] \Rightarrow 0=8$, which means no solutions.

b. (2 points) For which value(s) of $a$ (if any) does this system have **infinitely many solutions**? Briefly explain your answer.

None, b/c if $a \neq 3$, then $a-3 \neq 0$, so we can reduce the last row to:

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & \frac{a+5}{a-3} \end{array}\right]$$

each unknown has a leading one → unique solution

c. (3 points) Using $a=1$, finish solving the system. If it has infinitely many solutions, write the solution as a parameterized set.

$a=1$: ③ $\left[\begin{array}{ccc|c} 0 & 0 & -2 & 6 \end{array}\right]$ $\quad -2z=6 \Rightarrow \boxed{z=-3}$

$y-2z=5 \Rightarrow y=5+2(-3)=-1.$ $\boxed{y=-1}$

$x+4z=3 \Rightarrow x=3-4(-3)=15$ $\boxed{x=15}$

10. (6 points) Solve the following linear system by using Gauss-Jordan Elimination (i.e. finding the reduced row echelon form). If it has infinitely many solutions, provide an explicit parameterized form for these solutions. Show all work.

(Hint: As a checkpoint for your work, this system has infinitely many solutions).

$$\begin{array}{rcrcrcr} x & + & y & + & z & = & 5 \\ 2x & - & 3y & + & 12z & = & 5 \\ 3x & + & 2y & + & 5z & = & 14 \end{array}.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & -3 & 12 & 5 \\ 3 & 2 & 5 & 14 \end{array}\right] \xrightarrow[-3r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & -5 & 10 & -5 \\ 0 & -1 & 2 & -1 \end{array}\right] \xrightarrow{-\frac{1}{5}r_2\to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & -2 & 1 \\ 0 & -1 & 2 & -1 \end{array}\right]$$

$$\xrightarrow{r_2+r_3\to r_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-r_2+r_1\to r_1} \begin{array}{c} \begin{array}{cccc} x & y & z & \end{array} \\ \left[\begin{array}{ccc|c} 1 & 0 & 3 & 4 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

$$\boxed{\begin{array}{l} z = t \\ y = 1 + 2t \\ x = 4 - 3t \end{array}}$$